



LabCorp Dublin
6370 Wilcox Road
Dublin, OH 43016-1269

Phone: 800-282-7300

Specimen Number 252-127-9500-0		Patient ID		Control Number	Account Number 90000999	Account Phone Number 336-436-8645	Route 00
SAMPLE REPORT				Account Address LabCorp Test Master Test Account 3060 South Church Street Burlington NC 27215			
Patient First Name 500115		Patient Middle Name					
Patient SS#		Patient Phone		Total Volume			
Age (Y/M/D) 24/08/06	Date of Birth 01/02/92	Sex M	Fasting				
Patient Address				Additional Information NORMAL REPORT			
Date and Time Collected 09/08/16 00:00	Date Entered 09/08/16	Date and Time Reported		Physician Name	NPI	Physician ID	

Tests Ordered						
GlycoMark(R) (1,5 AG)						

TESTS	RESULT	FLAG	UNITS	REFERENCE INTERVAL	LAB
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GlycoMark(R) (1,5 AG)

GlycoMark(R) (1,5 AG) 24.5 ug/mL 01

GlycoMark(TM) is intended for use with managing glycemic control in diabetic patients. A low result corresponds to high glucose peaks.

1, 5-AG blood levels can be affected by clinical conditions or medications. Please refer to the directory of services or labcorp website test menu for detailed list of limitations.

Reference Range:

Adults Males: 10.7 - 32.0

Glycemic control goal for diabetic patients: >10

01	ES	Esoterix Endocrinology	Dir: Samuel Pepkowitz, MD
		4301 Lost Hills Road, Calabasas Hills, CA 91301-5358	
For inquiries, the physician may contact Branch: 800-222-7566 Lab: 800-282-7300			

SAMPLE REPORT, 500115		252-127-9500-0	Seq # 0000
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09/08/16 13:45 ET

DUPLICATE FINAL REPORT

Page 1 of 1

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DOC1 Ver: 1.49



LabCorp Dublin
6370 Wilcox Road
Dublin, OH 43016-1269

Phone: 800-282-7300

Specimen Number 252-127-9501-0		Patient ID		Control Number	Account Number 90000999	Account Phone Number 336-436-8645	Route 00
SAMPLE REPORT				Account Address LabCorp Test Master Test Account 3060 South Church Street Burlington NC 27215			
Patient First Name 500115		Patient Middle Name					
Patient SS#		Patient Phone		Total Volume			
Age (Y/M/D) 24/08/06	Date of Birth 01/02/92	Sex M	Fasting				
Patient Address				Additional Information ABNORMAL REPORT			
Date and Time Collected 09/08/16 00:00	Date Entered 09/08/16	Date and Time Reported		Physician Name	NPI	Physician ID	

Tests Ordered						
GlycoMark(R) (1,5 AG)						

TESTS	RESULT	FLAG	UNITS	REFERENCE INTERVAL	LAB
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GlycoMark(R) (1,5 AG)

GlycoMark(R) (1,5 AG)	1.8	Low	ug/mL		01
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GlycoMark(TM) is intended for use with managing glycemic control in diabetic patients. A low result corresponds to high glucose peaks.

1, 5-AG blood levels can be affected by clinical conditions or medications. Please refer to the directory of services or labcorp website test menu for detailed list of limitations.

Reference Range:

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SAMPLE REPORT, 500115		252-127-9501-0	Seq # 0000
------------------------------	--	-----------------------	------------

Name: Sample Report - Food Safe Basic
Lab ID:

Order #:
Pick Ticket #:
Date of Birth:

Report Date: 3/9/2017
Date Collected: 3/3/2017
Reference #: 409

FINAL REPORT

RESULT		- REFERENCE RANGE -			ALLERGEN	- NOT SAFE -		
		LOW	MODERATE	AVOID		LOW	MODERATE	AVOID
213	Avoid	<70	70 - 100	>100	CANDIDA SCREEN Candida albicans			
DAIRY								
376	Avoid	<100	100 - 350	>350	Casein			
268	Avoid	<100	100 - 250	>250	Cheddar Cheese			
636	Avoid	<100	100 - 350	>350	Cottage Cheese			
824	Avoid	<140	140 - 350	>350	Cow's Milk			
192	Moderate	<120	120 - 370	>370	Goat's Milk			
289	Moderate	<100	100 - 350	>350	Mozzarella Cheese			
1357	Avoid	<150	150 - 250	>250	Whey			
FISH								
124	Low	<200	200 - 350	>350	Cod			
70	Low	<190	190 - 340	>340	Halibut			
87	Low	<200	200 - 350	>350	Salmon			
70	Low	<150	150 - 300	>300	Sardine			
87	Low	<160	160 - 310	>310	Sole			
89	Low	<250	250 - 400	>400	Tilapia			
85	Low	<200	200 - 350	>350	Trout			
70	Low	<150	150 - 300	>300	Tuna			
FRUITS								
101	Low	<160	160 - 310	>310	Apple Mix			
72	Low	<120	120 - 270	>270	Apricot			
459	Avoid	<150	150 - 300	>300	Avocado			
82	Low	<160	160 - 340	>340	Banana			
99	Low	<130	130 - 280	>280	Blueberry			
69	Low	<100	100 - 250	>250	Cranberry			
96	Low	<150	150 - 300	>300	Grapefruit			
75	Low	<130	130 - 280	>280	Lemon			
115	Low	<130	130 - 280	>280	Orange			
174	Low	<180	180 - 310	>310	Papaya			
73	Low	<120	120 - 270	>270	Peach			
70	Low	<100	100 - 250	>250	Pear			
343	Avoid	<110	110 - 300	>300	Pineapple			
71	Low	<130	130 - 280	>280	Plum			
259	Moderate	<200	200 - 350	>350	Raspberry			
133	Low	<140	140 - 300	>300	Red Grape			
86	Low	<120	120 - 270	>270	Strawberry			
146	Low	<180	180 - 330	>330	Watermelon			
GRAINS								
221	Moderate	<150	150 - 300	>300	Barley			
101	Low	<150	150 - 300	>300	Buckwheat			
112	Low	<180	180 - 320	>320	Corn			
708	Avoid	<150	150 - 300	>300	Gliadin			
491	Avoid	<130	130 - 280	>280	Gluten			
109	Low	<180	180 - 330	>330	Hemp			
63	Low	<100	100 - 250	>250	Oat			
89	Low	<150	150 - 300	>300	Quinoa			
95	Low	<200	200 - 350	>350	Rice			
98	Low	<150	150 - 330	>330	Teff			
615	Avoid	<120	120 - 260	>260	Wheat			

Name: Order #: Report Date: 3/9/2017
Lab ID: Pick Ticket #: Date Collected: 3/3/2017
Date of Birth: Reference #: 409

FINAL REPORT

RESULT		- REFERENCE RANGE -			ALLERGEN	- NOT SAFE -		
		LOW	MODERATE	AVOID		LOW	MODERATE	AVOID
MEATS								
246	Moderate	<110	110 - 260	>260	Beef			
202	Moderate	<130	130 - 220	>220	Buffalo			
162	Moderate	<160	160 - 310	>310	Chicken			
1889	Avoid	<180	180 - 450	>450	Egg White			
1836	Avoid	<190	190 - 500	>500	Egg Yolk			
198	Moderate	<120	120 - 270	>270	Lamb			
502	Avoid	<150	150 - 300	>300	Pork			
105	Low	<150	150 - 300	>300	Turkey			
MISCELLANEOUS								
79	Low	<150	150 - 300	>300	Baker's Yeast			
77	Low	<100	100 - 250	>250	Brewer's Yeast			
87	Low	<200	200 - 350	>350	Cane Sugar			
116	Low	<130	130 - 280	>280	Coffee			
110	Low	<150	150 - 300	>300	Honey			
229	Moderate	<150	150 - 300	>300	Mushroom			
72	Low	<100	100 - 250	>250	Xanthan Gum			
NUTS								
130	Low	<130	130 - 280	>280	Almond			
327	Avoid	<150	150 - 300	>300	Peanut			
101	Low	<180	180 - 330	>330	Pecan			
109	Low	<150	150 - 300	>300	Pumpkin Seed			
109	Low	<200	200 - 350	>350	Sesame Seed			
132	Low	<180	180 - 320	>320	Sunflower Seed			
113	Low	<250	250 - 400	>400	Walnut			
SHELLFISH								
61	Low	<140	140 - 290	>290	Clam			
96	Low	<140	140 - 290	>290	Crab			
93	Low	<170	170 - 320	>320	Lobster			
141	Moderate	<110	110 - 260	>260	Oyster			
85	Low	<150	150 - 300	>300	Shrimp			
VEGETABLES								
92	Low	<120	120 - 270	>270	Asparagus			
88	Low	<140	140 - 290	>290	Beet			
95	Low	<150	150 - 300	>300	Black Olive			
79	Low	<140	140 - 290	>290	Broccoli			
90	Low	<120	120 - 270	>270	Cabbage			
71	Low	<140	140 - 290	>290	Carrot			
103	Low	<150	150 - 300	>300	Cauliflower			
90	Low	<140	140 - 290	>290	Celery			
108	Low	<150	150 - 300	>300	Cucumber			
799	Avoid	<180	180 - 380	>380	Garlic			
491	Avoid	<140	140 - 300	>300	Green Bean			
114	Low	<150	150 - 300	>300	Green Pepper			
1361	Avoid	<180	180 - 480	>480	Kidney Bean			
73	Low	<130	130 - 280	>280	Lentil			
106	Low	<150	150 - 300	>300	Lettuce			
178	Moderate	<140	140 - 300	>300	Lima Bean			
150	Moderate	<130	130 - 280	>280	Onion			
81	Low	<120	120 - 270	>270	Pea			
145	Low	<200	200 - 350	>350	Potato			
180	Moderate	<140	140 - 290	>290	Pumpkin			
103	Low	<150	150 - 300	>300	Soybean			
118	Low	<160	160 - 310	>310	Spinach			
86	Low	<150	150 - 300	>300	Tomato			

Name:

Reference #: **409**

PERSONALIZED FOUR DAY ROTATION FOOD PLAN - DAY 1 CHOICES

PROTEINS	DAIRY	GRAINS / FLOUR	LEGUMES	VEGETABLES	FRUITS	NUTS, SEEDS AND OILS	HERBS AND SPICES	MISCELLANEOUS
cod LAMB OYSTER	GOAT MILK oat milk	BARLEY oat		black olive carrot celery	banana papaya red grape			baker's yeast brewer's yeast cane sugar

PERSONALIZED FOUR DAY ROTATION FOOD PLAN - DAY 2 CHOICES

PROTEINS	DAIRY	GRAINS / FLOUR	LEGUMES	VEGETABLES	FRUITS	NUTS, SEEDS AND OILS	HERBS AND SPICES	MISCELLANEOUS
crab lobster salmon trout turkey	almond milk potato milk	potato flour potato starch		beet green pepper MUSHROOM potato spinach tomato	apple mix blueberry cranberry pear	almond		apple used as sweetener MUSHROOM pear used as sweetener

PERSONALIZED FOUR DAY ROTATION FOOD PLAN - DAY 3 CHOICES

PROTEINS	DAIRY	GRAINS / FLOUR	LEGUMES	VEGETABLES	FRUITS	NUTS, SEEDS AND OILS	HERBS AND SPICES	MISCELLANEOUS
BEEF clam halibut sole tofu	MOZZARELLA CHEESE soy milk	buckwheat corn	lentil LIMA BEAN soybean	asparagus corn cucumber ONION pea	RASPBERRY strawberry watermelon	pecan soy oil walnut walnut oil		coffee honey

PERSONALIZED FOUR DAY ROTATION FOOD PLAN - DAY 4 CHOICES

PROTEINS	DAIRY	GRAINS / FLOUR	LEGUMES	VEGETABLES	FRUITS	NUTS, SEEDS AND OILS	HERBS AND SPICES	MISCELLANEOUS
CHICKEN sardine shrimp tuna	rice milk sesame seed milk	rice		broccoli cabbage cauliflower lettuce	apricot grapefruit lemon orange peach plum	sesame oil sesame seed sunflower oil sunflower seed		

KEY: low reaction = lowercase

MODERATE REACTION = UPPERCASE

SIGNIFICANT = IS NOT LISTED IN ROTATION

SUMMARY OF FOOD

LOW REACTION FOODS

Almond	Apple Mix	Apricot	Asparagus	Baker's Yeast	Banana	Beet	Black Olive	Blueberry
Brewer's Yeast	Broccoli	Buckwheat	Cabbage	Cane Sugar	Carrot	Cauliflower	Celery	Clam
Cod	Coffee	Corn	Crab	Cranberry	Cucumber	Grapefruit	Green Pepper	Halibut
Hemp	Honey	Lemon	Lentil	Lettuce	Lobster	Oat	Orange	Papaya
Pea	Peach	Pear	Pecan	Plum	Potato	Pumpkin Seed	Quinoa	Red Grape
Rice	Salmon	Sardine	Sesame Seed	Shrimp	Sole	Soybean	Spinach	Strawberry
Sunflower Seed	Teff	Tilapia	Tomato	Trout	Tuna	Turkey	Walnut	Watermelon
Xanthan Gum								

MODERATE CONSUMPTION - MAY EAT ONCE EVERY FOUR DAYS

Barley	Beef	Buffalo	Chicken	Goat's Milk	Lamb	Lima Bean	Mozzarella Cheese	Mushroom
Onion	Oyster	Pumpkin	Raspberry					

AVOID THESE FOODS

Avocado	Candida albicans	Casein	Cheddar Cheese	Cottage Cheese	Cow's Milk	Egg White	Egg Yolk	Garlic
Gliadin	Gluten	Green Bean	Kidney Bean	Peanut	Pineapple	Pork	Wheat	Whey



LAB #: F000000-0000-0
PATIENT: Sample Patient
ID: P0000000000
SEX: Female
DOB:

AGE: 71

CLIENT #: 12345
DOCTOR:
Doctor's Data, Inc.
3755 Illinois Ave.
St. Charles, IL 60174 U.S.A.

Comprehensive Stool Analysis / Parasitology x1

BACTERIOLOGY CULTURE

Expected/Beneficial flora

4+ Bacteroides fragilis group
3+ Bifidobacterium spp.
4+ Escherichia coli
3+ Lactobacillus spp.
4+ Enterococcus spp.

2+ Clostridium spp.
NG = No Growth

Commensal (Imbalanced) flora

4+ Alpha hemolytic strep
2+ Hemolytic Escherichia coli
1+ Proteus mirabilis

Dysbiotic flora

4+ Citrobacter koseri
4+ Enterobacter cloacae complex
4+ Klebsiella oxytoca
4+ Klebsiella pneumoniae ssp pneumoniae
2+ Possible Salmonella spp
Sent to State for Confirmation

BACTERIA INFORMATION

Expected /Beneficial bacteria make up a significant portion of the total microflora in a healthy & balanced GI tract. These beneficial bacteria have many health-protecting effects in the GI tract including manufacturing vitamins, fermenting fibers, digesting proteins and carbohydrates, and propagating anti-tumor and anti-inflammatory factors.

Clostridia are prevalent flora in a healthy intestine. Clostridium spp. should be considered in the context of balance with other expected/beneficial flora. Absence of clostridia or over abundance relative to other expected/beneficial flora indicates bacterial imbalance. If *C. difficile* associated disease is suspected, a Comprehensive Clostridium culture or toxigenic *C. difficile* DNA test is recommended.

Commensal (Imbalanced) bacteria are usually neither pathogenic nor beneficial to the host GI tract. Imbalances can occur when there are insufficient levels of beneficial bacteria and increased levels of commensal bacteria. Certain commensal bacteria are reported as dysbiotic at higher levels.

Dysbiotic bacteria consist of known pathogenic bacteria and those that have the potential to cause disease in the GI tract. They can be present due to a number of factors including: consumption of contaminated water or food, exposure to chemicals that are toxic to beneficial bacteria; the use of antibiotics, oral contraceptives or other medications; poor fiber intake and high stress levels.

YEAST CULTURE

Normal flora

1+ Candida albicans
1+ Candida parapsilosis

Dysbiotic flora

MICROSCOPIC YEAST

Result:

Few

Expected:

None - Rare

The microscopic finding of yeast in the stool is helpful in identifying whether there is proliferation of yeast. Rare yeast may be normal; however, yeast observed in higher amounts (few, moderate, or many) is abnormal.

YEAST INFORMATION

Yeast normally can be found in small quantities in the skin, mouth, intestine and mucocutaneous junctions. Overgrowth of yeast can infect virtually every organ system, leading to an extensive array of clinical manifestations. Fungal diarrhea is associated with broad-spectrum antibiotics or alterations of the patient's immune status. Symptoms may include abdominal pain, cramping and irritation. When investigating the presence of yeast, disparity may exist between culturing and microscopic examination. Yeast are not uniformly dispersed throughout the stool, this may lead to undetectable or low levels of yeast identified by microscopy, despite a cultured amount of yeast. Conversely, microscopic examination may reveal a significant amount of yeast present, but no yeast cultured. Yeast does not always survive transit through the intestines rendering it unviable.

Comments:

Date Collected: 02/27/2015

Date Received: 03/02/2015

Date Completed: 03/10/2015

* *Aeromonas, Campylobacter, Plesiomonas, Salmonella, Shigella, Vibrio, Yersinia, & Edwardsiella tarda* have been specifically tested for and found absent unless reported.





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Comprehensive Stool Analysis / Parasitology x1

PARASITOLOGY/MICROSCOPY *

Sample 1

None Ova or Parasites
Few Yeast

*A trichrome stain and concentrated iodine wet mount slide is read for each sample submitted.

PARASITOLOGY INFORMATION

Intestinal parasites are abnormal inhabitants of the gastrointestinal tract that have the potential to cause damage to their host. The presence of any parasite within the intestine generally confirms that the patient has acquired the organism through fecal-oral contamination. Damage to the host includes parasitic burden, migration, blockage and pressure. Immunologic inflammation, hypersensitivity reactions and cytotoxicity also play a large role in the morbidity of these diseases. The infective dose often relates to severity of the disease and repeat encounters can be additive.

There are two main classes of intestinal parasites, they include protozoa and helminths. The protozoa typically have two stages; the trophozoite stage that is the metabolically active, invasive stage and the cyst stage, which is the vegetative inactive form resistant to unfavorable environmental conditions outside the human host. Helminths are large, multicellular organisms. Like protozoa, helminths can be either free-living or parasitic in nature. In their adult form, helminths cannot multiply in humans.

In general, acute manifestations of parasitic infection may involve diarrhea with or without mucus and or blood, fever, nausea, or abdominal pain. However these symptoms do not always occur. Consequently, parasitic infections may not be diagnosed or eradicated. If left untreated, chronic parasitic infections can cause damage to the intestinal lining and can be an unsuspected cause of illness and fatigue. Chronic parasitic infections can also be associated with increased intestinal permeability, irritable bowel syndrome, irregular bowel movements, malabsorption, gastritis or indigestion, skin disorders, joint pain, allergic reactions, and decreased immune function.

In some instances, parasites may enter the circulation and travel to various organs causing severe organ diseases such as liver abscesses and cysticercosis. In addition, some larval migration can cause pneumonia and in rare cases hyper infection syndrome with large numbers of larvae being produced and found in every tissue of the body.

One negative parasitology x1 specimen does not rule out the possibility of parasitic disease, parasitology x3 is recommended. This exam is not designed to detect *Cryptosporidium* spp, *Cyclospora cayentanensis* or *Microsporidia* spp.

GIARDIA/CRYPTOSPORIDIUM IMMUNOASSAY

	Within	Outside	Reference Range
Giardia intestinalis	Neg		Neg
Cryptosporidium	Neg		Neg

Giardia intestinalis (lamblia) is a protozoan that infects the small intestine and is passed in stool and spread by the fecal-oral route. Waterborne transmission is the major source of giardiasis.

Cryptosporidium is a coccidian protozoa that can be spread from direct person-to-person contact or waterborne transmission.

Comments:

Date Collected: 02/27/2015

Date Received: 03/02/2015

Date Completed: 03/10/2015



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Comprehensive Stool Analysis / Parasitology x1

DIGESTION / ABSORPTION

	Within	Outside	Reference Range
Elastase	410		> 200 µg/mL
Fat Stain	None		None - Mod
Muscle fibers	Rare		None - Rare
Vegetable fibers	Rare		None - Few
Carbohydrates	Neg		Neg

Elastase findings can be used for the diagnosis or the exclusion of exocrine pancreatic insufficiency. Correlations between low levels and chronic pancreatitis and cancer have been reported. **Fat Stain:** Microscopic determination of fecal fat using Sudan IV staining is a qualitative procedure utilized to assess fat absorption and to detect steatorrhea. **Muscle fibers** in the stool are an indicator of incomplete digestion. Bloating, flatulence, feelings of "fullness" may be associated with increase in muscle fibers. **Vegetable fibers** in the stool may be indicative of inadequate chewing, or eating "on the run". **Carbohydrates:** The presence of reducing substances in stool specimens can indicate carbohydrate malabsorption.

INFLAMMATION

	Within	Outside	Reference Range
Lactoferrin		8.0	< 7.3 µg/mL
Calprotectin*		85	<= 50 µg/g
Lysozyme*	78		<= 600 ng/mL
White Blood Cells	None		None - Rare
Mucus	Neg		Neg

Lactoferrin and **Calprotectin** are reliable markers for differentiating organic inflammation (IBD) from functional symptoms (IBS) and for management of IBD. Monitoring levels of fecal lactoferrin and calprotectin can play an essential role in determining the effectiveness of therapy, are good predictors of IBD remission, and can indicate a low risk of relapse. **Lysozyme*** is an enzyme secreted at the site of inflammation in the GI tract and elevated levels have been identified in IBD patients. **White Blood Cells** (WBC) and **Mucus** in the stool can occur with bacterial and parasitic infections, with mucosal irritation, and inflammatory bowel diseases such as Crohn's disease or ulcerative colitis.

IMMUNOLOGY

	Within	Outside	Reference Range
Secretory IgA*		288	51 - 204 mg/dL

Secretory IgA* (sIgA) is secreted by mucosal tissue and represents the first line of defense of the GI mucosa and is central to the normal function of the GI tract as an immune barrier. Elevated levels of sIgA have been associated with an upregulated immune response.

Comments:

Date Collected: 02/27/2015
Date Received: 03/02/2015
Date Completed: 03/10/2015

*For Research Use Only. Not for use in diagnostic procedures.

Methodology: Elisa, Microscopy, Colormetric,
Gas Chromatography, pH Electrode



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Comprehensive Stool Analysis / Parasitology x1

SHORT CHAIN FATTY ACIDS

	Within	Outside	Reference Range
% Acetate	61		40 - 75 %
% Propionate		30	9 - 29 %
% Butyrate	9.6		9 - 37 %
% Valerate	0.5		0.5 - 7 %
Butyrate		0.52	0.8 - 4.8 mg/mL
Total SCFA's	5.5		4 - 18 mg/mL

Short chain fatty acids (SCFAs): SCFAs are the end product of the bacterial fermentation process of dietary fiber by beneficial flora in the gut and play an important role in the health of the GI as well as protecting against intestinal dysbiosis. Lactobacilli and bifidobacteria produce large amounts of short chain fatty acids, which decrease the pH of the intestines and therefore make the environment unsuitable for pathogens, including bacteria and yeast. Studies have shown that SCFAs have numerous implications in maintaining gut physiology. SCFAs decrease inflammation, stimulate healing, and contribute to normal cell metabolism and differentiation. Levels of **Butyrate** and **Total SCFA** in mg/mL are important for assessing overall SCFA production, and are reflective of beneficial flora levels and/or adequate fiber intake.

INTESTINAL HEALTH MARKERS

	Within	Outside	Reference Range
Red Blood Cells	None		None - Rare
pH	6.6		6 - 7.8
Occult Blood	Neg		Neg

Red Blood Cells (RBC) in the stool may be associated with a parasitic or bacterial infection, or an inflammatory bowel condition such as ulcerative colitis. Colorectal cancer, anal fistulas, and hemorrhoids should also be ruled out.

pH: Fecal pH is largely dependent on the fermentation of fiber by the beneficial flora of the gut.

Occult blood: A positive occult blood indicates the presence of free hemoglobin found in the stool, which is released when red blood cells are lysed.

MACROSCOPIC APPEARANCE

	Appearance	Expected
Color	Brown	Brown
Consistency	Soft	Formed/Soft

Color: Stool is normally brown because of pigments formed by bacteria acting on bile introduced into the digestive system from the liver. While certain conditions can cause changes in stool color, many changes are harmless and are caused by pigments in foods or dietary supplements. **Consistency:** Stool normally contains about 75% water and ideally should be formed and soft. Stool consistency can vary based upon transit time and water absorption.

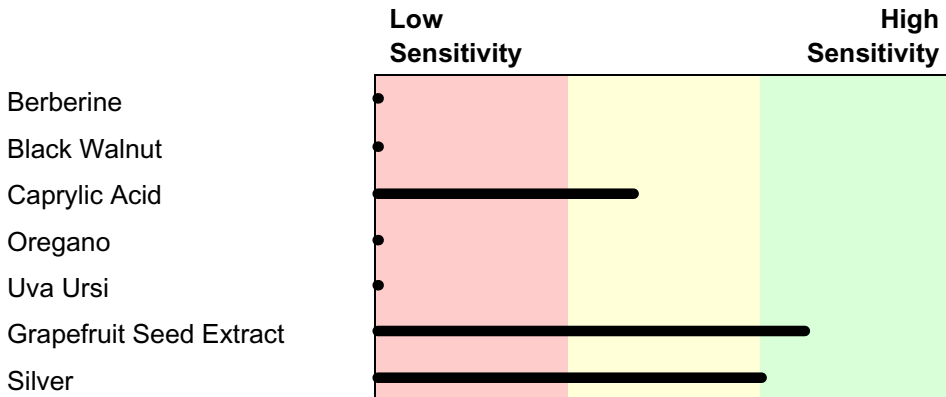


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Bacterial Susceptibilities: *Citrobacter koseri*

NATURAL ANTIBACTERIALS



Natural antibacterial agents may be useful for treatment of patients when organisms display in-vitro sensitivity to these agents. The test is performed by using standardized techniques and filter paper disks impregnated with the listed agent. Relative sensitivity is reported for each natural agent based upon the diameter of the zone of inhibition surrounding the disk. Data based on over 5000 individual observations were used to relate the zone size to the activity level of the agent. A scale of relative sensitivity is defined for the natural agents tested.

PRESCRIPTIVE AGENTS

	Resistant	Intermediate	Susceptible
Amoxicillin-Clavulanic Acid			S
Ampicillin	R		
Cefazolin			S
Ceftazidime			S
Ciprofloxacin			S
Trimeth-sulfa			S

Susceptible results imply that an infection due to the bacteria may be appropriately treated when the recommended dosage of the tested antimicrobial agent is used.

Intermediate results imply that response rates may be lower than for susceptible bacteria when the tested antimicrobial agent is used.

Resistant results imply that the bacteria will not be inhibited by normal dosage levels of the tested antimicrobial agent.

Comments:

Date Collected: 02/27/2015

Date Received: 03/02/2015

Date Completed: 03/10/2015

Natural antibacterial agent susceptibility testing is intended for research use only.

Not for use in diagnostic procedures.

v10.11

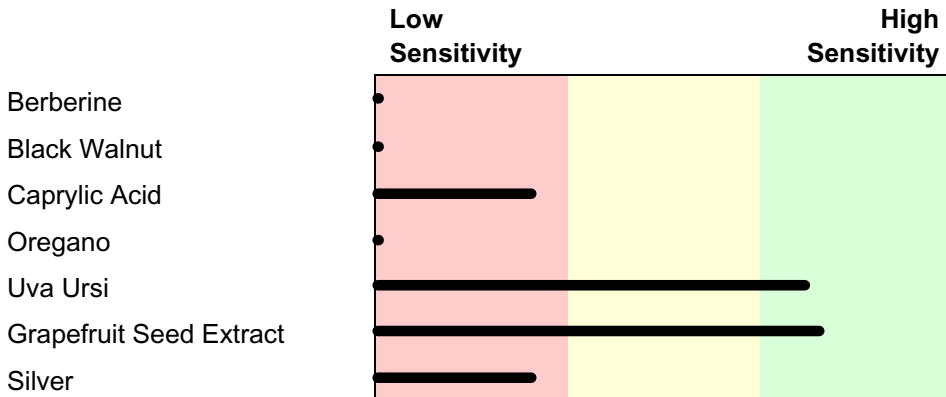


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Bacterial Susceptibilities: Enterobacter cloacae complex

NATURAL ANTIBACTERIALS



Natural antibacterial agents may be useful for treatment of patients when organisms display in-vitro sensitivity to these agents. The test is performed by using standardized techniques and filter paper disks impregnated with the listed agent. Relative sensitivity is reported for each natural agent based upon the diameter of the zone of inhibition surrounding the disk. Data based on over 5000 individual observations were used to relate the zone size to the activity level of the agent. A scale of relative sensitivity is defined for the natural agents tested.

PRESCRIPTIVE AGENTS

	Resistant	Intermediate	Susceptible
Amoxicillin-Clavulanic Acid	R		
Ampicillin	R		
Cefazolin	R		
Ceftazidime			S
Ciprofloxacin			S
Trimeth-sulfa			S

Susceptible results imply that an infection due to the bacteria may be appropriately treated when the recommended dosage of the tested antimicrobial agent is used.

Intermediate results imply that response rates may be lower than for susceptible bacteria when the tested antimicrobial agent is used.

Resistant results imply that the bacteria will not be inhibited by normal dosage levels of the tested antimicrobial agent.

Comments:

Date Collected: 02/27/2015

Date Received: 03/02/2015

Date Completed: 03/10/2015

Natural antibacterial agent susceptibility testing is intended for research use only.

Not for use in diagnostic procedures.

v10.11

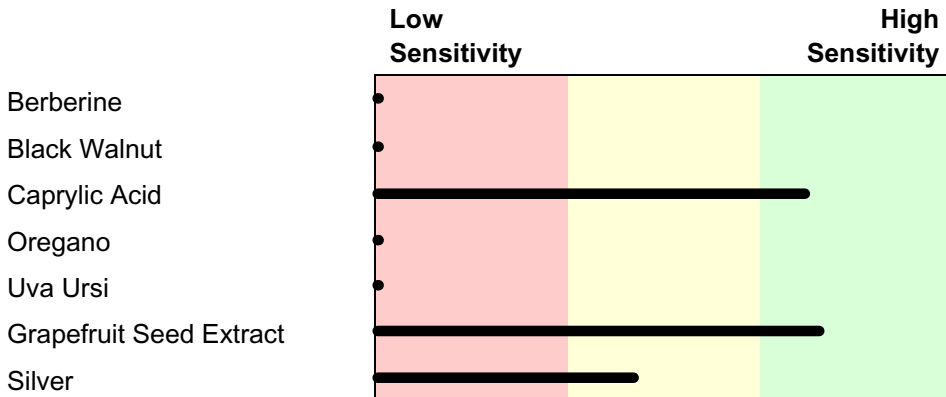


LAB #: F000000-0000-0
PATIENT: Sample Patient
ID: P0000000000
SEX: Female
AGE: 71

CLIENT #: 12345
DOCTOR:
Doctor's Data, Inc.
3755 Illinois Ave.
St. Charles, IL 60174 U.S.A.

Bacterial Susceptibilities: *Klebsiella oxytoca*

NATURAL ANTIBACTERIALS



Natural antibacterial agents may be useful for treatment of patients when organisms display in-vitro sensitivity to these agents. The test is performed by using standardized techniques and filter paper disks impregnated with the listed agent. Relative sensitivity is reported for each natural agent based upon the diameter of the zone of inhibition surrounding the disk. Data based on over 5000 individual observations were used to relate the zone size to the activity level of the agent. A scale of relative sensitivity is defined for the natural agents tested.

PRESCRIPTIVE AGENTS

	Resistant	Intermediate	Susceptible
Amoxicillin-Clavulanic Acid			S
Ampicillin	R		
Cefazolin	R		
Ceftazidime			S
Ciprofloxacin			S
Trimeth-sulfa			S

Susceptible results imply that an infection due to the bacteria may be appropriately treated when the recommended dosage of the tested antimicrobial agent is used.

Intermediate results imply that response rates may be lower than for susceptible bacteria when the tested antimicrobial agent is used.

Resistant results imply that the bacteria will not be inhibited by normal dosage levels of the tested antimicrobial agent.

Comments:

Date Collected: 02/27/2015

Date Received: 03/02/2015

Date Completed: 03/10/2015

Natural antibacterial agent susceptibility testing is intended for research use only.

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v10.11

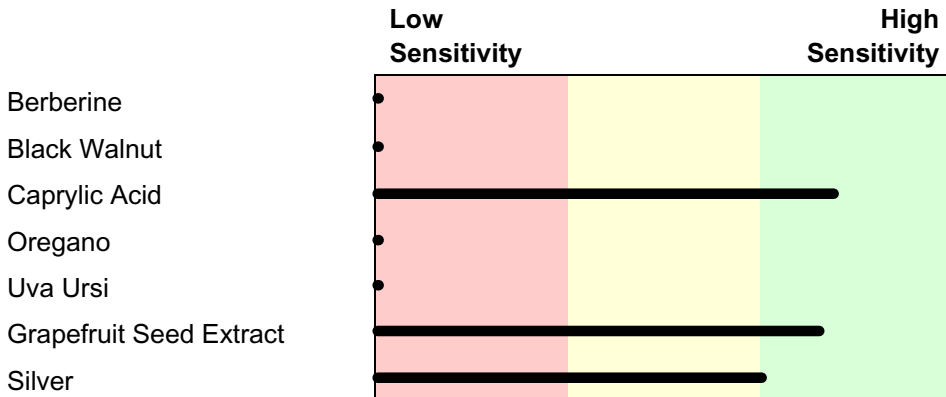


LAB #: F000000-0000-0
PATIENT: Sample Patient
ID: P0000000000
SEX: Female
AGE: 71

CLIENT #: 12345
DOCTOR:
Doctor's Data, Inc.
3755 Illinois Ave.
St. Charles, IL 60174 U.S.A.

Bacterial Susceptibilities: *Klebsiella pneumoniae* ssp *pneumoniae*

NATURAL ANTIBACTERIALS



Natural antibacterial agents may be useful for treatment of patients when organisms display in-vitro sensitivity to these agents. The test is performed by using standardized techniques and filter paper disks impregnated with the listed agent. Relative sensitivity is reported for each natural agent based upon the diameter of the zone of inhibition surrounding the disk. Data based on over 5000 individual observations were used to relate the zone size to the activity level of the agent. A scale of relative sensitivity is defined for the natural agents tested.

PRESCRIPTIVE AGENTS

	Resistant	Intermediate	Susceptible
Amoxicillin-Clavulanic Acid			S
Ampicillin	R		
Cefazolin			S
Ceftazidime			S
Ciprofloxacin			S
Trimeth-sulfa			S

Susceptible results imply that an infection due to the bacteria may be appropriately treated when the recommended dosage of the tested antimicrobial agent is used.

Intermediate results imply that response rates may be lower than for susceptible bacteria when the tested antimicrobial agent is used.

Resistant results imply that the bacteria will not be inhibited by normal dosage levels of the tested antimicrobial agent.

Comments:

Date Collected: 02/27/2015

Date Received: 03/02/2015

Date Completed: 03/10/2015

Natural antibacterial agent susceptibility testing is intended for research use only.

Not for use in diagnostic procedures.

v10.11

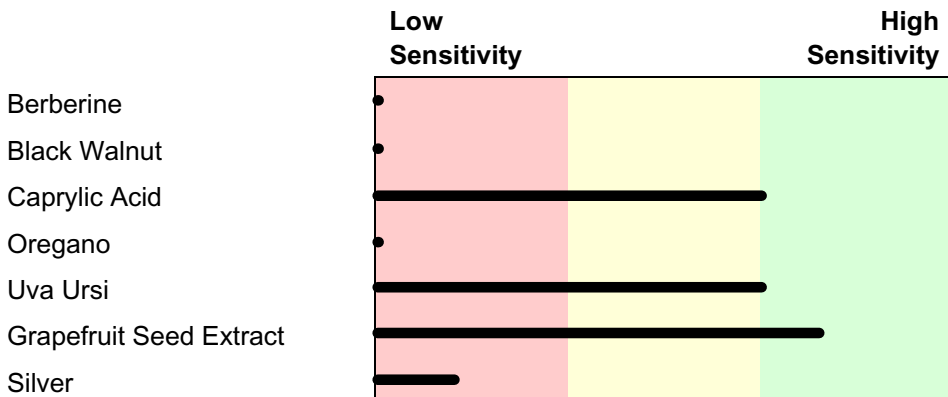


LAB #: F000000-0000-0
PATIENT: Sample Patient
ID: P0000000000
SEX: Female
AGE: 71

CLIENT #: 12345
DOCTOR:
Doctor's Data, Inc.
3755 Illinois Ave.
St. Charles, IL 60174 U.S.A.

Bacterial Susceptibilities: Possible Salmonella spp

NATURAL ANTIBACTERIALS



Natural antibacterial agents may be useful for treatment of patients when organisms display in-vitro sensitivity to these agents. The test is performed by using standardized techniques and filter paper disks impregnated with the listed agent. Relative sensitivity is reported for each natural agent based upon the diameter of the zone of inhibition surrounding the disk. Data based on over 5000 individual observations were used to relate the zone size to the activity level of the agent. A scale of relative sensitivity is defined for the natural agents tested.

PRESCRIPTIVE AGENTS

	Resistant	Intermediate	Susceptible
Ampicillin			S
Ciprofloxacin			S
Trimeth-sulfa			S

Susceptible results imply that an infection due to the bacteria may be appropriately treated when the recommended dosage of the tested antimicrobial agent is used.

Intermediate results imply that response rates may be lower than for susceptible bacteria when the tested antimicrobial agent is used.

Resistant results imply that the bacteria will not be inhibited by normal dosage levels of the tested antimicrobial agent.

Comments:

Date Collected: 02/27/2015

Date Received: 03/02/2015

Date Completed: 03/10/2015

Natural antibacterial agent susceptibility testing is intended for research use only.

Not for use in diagnostic procedures.

v10.11



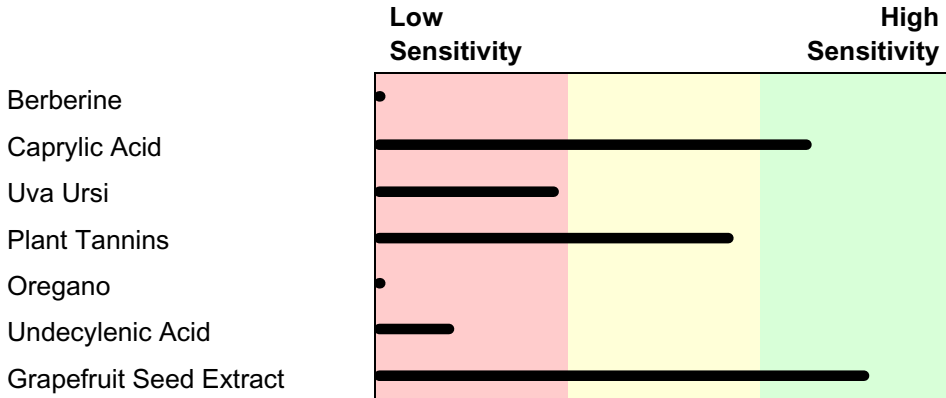


LAB #: F000000-0000-0
PATIENT: Sample Patient
ID: P0000000000
SEX: Female
AGE: 71

CLIENT #: 12345
DOCTOR:
Doctor's Data, Inc.
3755 Illinois Ave.
St. Charles, IL 60174 U.S.A.

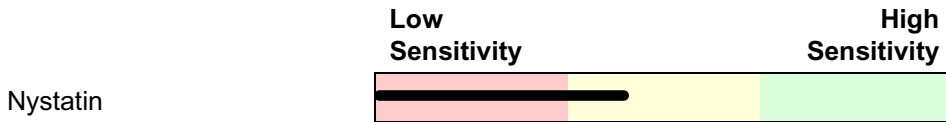
Yeast Susceptibilities: *Candida parapsilosis*

NATURAL ANTIFUNGALS



Natural antifungal agents may be useful for treatment of patients when organisms display in-vitro sensitivity to these agents. The test is performed by using standardized techniques and filter paper disks impregnated with the listed agent. Relative sensitivity is reported for each natural agent based upon the diameter of the zone of inhibition surrounding the disk. Data based on over 5000 individual observations were used to relate the zone size to the activity level of the agent. A scale of relative sensitivity is defined for the natural agents tested.

NON-ABSORBED ANTIFUNGALS



Non-absorbed antifungals may be useful for treatment of patients when organisms display in-vitro sensitivity to these agents. The test is performed using standardized commercially prepared disks impregnated with Nystatin. Relative sensitivity is reported based upon the diameter of the zone of inhibition surrounding the disk.

AZOLE ANTIFUNGALS

	Resistant	S-DD	Susceptible
Fluconazole			S
Itraconazole			S
Ketoconazole			S

Susceptible results imply that an infection due to the fungus may be appropriately treated when the recommended dosage of the tested antifungal agent is used.

Susceptible - Dose Dependent (S-DD) results imply that an infection due to the fungus may be treated when the highest recommended dosage of the tested antifungal agent is used.

Resistant results imply that the fungus will not be inhibited by normal dosage levels of the tested antifungal agent.

Standardized test interpretive categories established for *Candida* spp. are used for all yeast isolates.

Comments:

Date Collected: 02/27/2015
Date Received: 03/02/2015
Date Completed: 03/10/2015

Yeast antifungal susceptibility testing is intended for research use only.
Not for use in diagnostic procedures.

v10.11

INTRODUCTION

This analysis of the stool specimen provides fundamental information about the overall gastrointestinal health of the patient. When abnormal microflora or significant aberrations in intestinal health markers are detected, specific interpretive paragraphs are presented. If no significant abnormalities are found, interpretive paragraphs are not presented.

Clostridium spp

Clostridia are expected inhabitants of the human intestine. Although most clostridia in the intestine are not virulent, certain species have been associated with disease. Clostridium perfringens is a major cause of food poisoning and is also one cause of antibiotic-associated diarrhea. Clostridium difficile is a causative agent in antibiotic-associated diarrhea and pseudomembranous colitis. Other species reported to be prevalent in high amounts in patients with Autistic Spectrum Disorder include Clostridium histolyticum group, Clostridium cluster I, Clostridium bolteae, and Clostridium tetani.

If these disease associations are a concern further testing may be necessary.

Washington W, Allen S, Janda W, Koneman E, Procop G, Schreckenberger P, Woods, G. Koneman's Color Atlas and Textbook of Diagnostic Microbiology, 6th edition. Lippincott Williams and Wilkins; 2006. pg 931-939

Song Y, Liu C, Finegold SM. Real-Time PCR Quantitation of Clostridia in Feces of Autistic Children. Applied and Environmental Microbiology. Nov. 2004, 6459-6465.

Parracho H, Bingham MO, Gibson GR, McCartney AL. Differences Between the Gut Microflora of Children with Autistic Spectrum Disorders and That of Healthy Children. Journal of Medical Microbiology. 2005;54, 987-991.

Imbalanced flora

Imbalanced flora are those bacteria that reside in the host gastrointestinal tract and neither injure nor benefit the host. Certain dysbiotic bacteria may appear under the imbalances category if found at low levels because they are not likely pathogenic at the levels detected. When imbalanced flora appear, it is not uncommon to find inadequate levels of one or more of the beneficial bacteria and/or a fecal pH which is more towards the alkaline end of the reference range (6 - 7.8). It is also not uncommon to find hemolytic or mucoid E. coli with a concomitant deficiency of beneficial E. coli and alkaline pH, secondary to a mutation of beneficial E. coli in alkaline conditions (DDI observations). Treatment with antimicrobial agents is unnecessary unless bacteria appear under the dysbiotic category.

Mackowiak PA. The normal microbial flora. N Engl J Med. 1982;307(2):83-93.

Dysbiotic Flora

In a healthy balanced state of intestinal flora, the beneficial bacteria make up a significant proportion of the total microflora. However, in many individuals there is an imbalance or deficiency of beneficial flora and an overgrowth of non-beneficial (imbalance) or even pathogenic microorganisms (dysbiosis). This can be due to a number of factors including: consumption of contaminated water or food; daily exposure of chemicals that are toxic to beneficial bacteria; the use of antibiotics, oral contraceptives or other medications; poor fiber intake and high stress levels.

A number of toxic substances can be produced by the dysbiotic bacteria including amines, ammonia, hydrogen sulfide, phenols, and secondary bile acids which may cause inflammation or damage to the brush border of the intestinal lining. If left unchecked, long-term damage to the intestinal lining may result in leaky gut syndrome, allergies, autoimmune disease (e.g. rheumatoid arthritis), irritable bowel syndrome, fatigue, chronic headaches, and sensitivities to a variety of foods. In addition, pathogenic bacteria can cause acute symptoms such as abdominal pain, nausea, diarrhea, vomiting, and fever in cases of food poisoning.

Bacterial sensitivities to a variety of prescriptive and natural agents have been provided for the pathogenic bacteria that were cultured from this patient's specimen. This provides the practitioner with useful information to help plan an appropriate treatment regimen. Supplementation with probiotics or consumption of foods (yogurt, kefir, miso, tempeh, tamari sauce) containing strains of lactobacilli, bifidobacteria, and enterococci can help restore healthy flora levels. Polyphenols in green and ginseng tea have been found to increase the numbers of beneficial bacteria. Hypochlorhydria may also predispose an individual to bacterial overgrowth, particularly in the small intestine. Nutritional anti-inflammatories can aid in reversing irritation to the GI lining. These include quercetin, vitamin C, curcumin, gamma-linoleic acid, omega-3 fatty acids (EPA, DHA), and aloe vera. Other nutrients such as zinc, beta-carotene, pantothenic acid, and L-glutamine provide support for regeneration of the GI mucosa. A comprehensive program may be helpful in individuals in whom a dysbiotic condition has caused extensive GI damage.

Lispi E. Digestive Wellness. New Canaan,CT: Keats Publishing;1996.

Mitsuoka T. Intestinal Flora and Aging. Nutr Rev 1992;50(12):438-446.

Weisburger JH. Tea and Health: The Underlying Mechanisms. Proc Soc Exp Biol Med 1999;220(4):271-275.4.

Pereira SP, Gainsborough N, Dowling RH. Drug-induced Hypochlorhydria Causes High Duodenal Bacterial Counts in the Elderly. Ailment Pharmacol Ther 1998;12(1)99-104.

Murray MT. Stomach Ailments and Digestive Disturbances. Rocklin, CA: Prima Publishing; 1997.

Citrobacter species

Citrobacter species, a gram-negative bacterium and member of the Enterobacteriaceae family, is considered dysbiotic at 3+ or greater.

Citrobacter freundii complex, including Citrobacter freundii, Citrobacter braakii, Citrobacter gullenii, Citrobacter murlinae, Citrobacter rodentium, Citrobacter werkmanii, Citrobacter oungae, and less commonly, Citrobacter koseri and Citrobacter farmeri, can cause diarrheal disease. Symptoms due to Citrobacter freundii complex seem to be a result of the elaboration of an E. coli-like heat-stable enterotoxin and hydrogen sulfide. Citrobacter freundii complex has been implicated as a cause of gastrointestinal infection and inflammation, acute dysentery, and dyspepsia. Acute symptoms can include profuse, watery diarrhea which is often unaccompanied by abdominal pain, fecal blood, or white blood cells.

Citrobacter species thrive on Fructooligosaccharides (FOS), a common ingredient in artificial or alternative sweetener.

Antibiotics may be indicated if symptoms are prolonged. Refer to the bacterial sensitivities to identify the most appropriate agent.

Guarino, A, et al. Production of Escherichia coli Sta-Like Heat-Stable Enterotoxin by Citrobacter freundii Isolated from Humans. Journal of Clinical Microbiology. January 1987;110-114.

Washington W, Allen S, Janda W, Koneman E, Procop G, Schreckenberger P, Woods, G. Koneman's Color Atlas and Textbook of Diagnostic Microbiology, 6th edition. Lippincott Williams and Wilkins; 2006. pg 686-689.

Murray PR, Baron EJ, Jorgensen JH, Pfaller MA, Tenover FC, Tenover FC. Manual of Clinical Microbiology, 8th edition. Washington, DC: ASM Press; 2003. pg 701-704.

Enterobacter cloacae

Enterobacter cloacae is part of the Enterobacteriaceae family. E. cloacae complex is a group of six closely related species with similar resistance patterns, E. cloacae, E. asburiae, E. hormaechei, E. kobei, E. ludwigii, and E. nimipressuralis. This gram-negative bacterium is considered dysbiotic at levels of 3+ or greater. E. cloacae complex is considered an opportunistic pathogen associated with diarrhea in children. A Shiga-like toxin-producing E. cloacae was isolated from the feces of an infant with hemolytic-uremic syndrome. However, E. cloacae is most often involved in extraintestinal infections including the urinary tract, respiratory tract, and cutaneous wounds.

Widely distributed in the environment, Enterobacter is commonly isolated from both human and animal feces. Environmental strains of Enterobacter are capable of growth in foods at refrigeration temperature.

E. cloacae complex is known to possess inducible β -lactamases. Isolates may become

resistant to all cephalosporins after initiation of therapy. Avoid β -lactam-inhibitor drugs such as: amoxicillin / clavulanate, ampicillin / sulbactam, and piperacillin / tazobactam. Antibiotics may be indicated if symptoms are prolonged and in systemic infections. Refer to the bacterial sensitivities to identify the most appropriate agent.

Paton AW, Paton JC. Enterobacter cloacae Producing a Shiga-Like Toxin II-Related Cytotoxin Associated with a Case of Hemolytic-Uremic Syndrome. J Clin Microbiol. 1996;1105-1109.

Paterson DL. Serious Infections Caused by Enteric Gram-Negative Bacilli-Mechanisms of Antibiotic Resistance and Implications for Therapy of Gram-Negative Sepsis in the Transplanted Patient. Semin Respir Infect. 2002;17(4):260-4.

Ronald A. The Etiology of Urinary Tract Infection: Traditional and Emerging Pathogens. Dis Mon. 2003;49(2):71 - 82.

Washington W, Allen S, Janda W, Koneman E, Procop G, Schreckenberger P, Woods, G. Koneman's Color Atlas and Textbook of Diagnostic Microbiology, 6th edition. Lippincott Williams and Wilkins; 2006. pg 264-265.

Klebsiella species

Klebsiella belongs to the Enterobacteriaceae family and is closely related to the genera Enterobacter and Serratia. This gram-negative bacterium is considered dysbiotic in the amount of 3 - 4+.

Klebsiellae are widely distributed in nature and in the gastrointestinal tract of humans. In humans, they may colonize the skin, oral cavity, pharynx, or gastrointestinal tract. Klebsiellae may be regarded as normal flora in many parts of the colon, intestinal tract and biliary tract, but the gut is also the main reservoir of opportunistic strains.

This bacterium has the potential to cause intestinal, lung, urinary tract, and wound infections in susceptible individuals, but Klebsiella overgrowth is commonly asymptomatic. K. pneumoniae, in particular, may cause diarrhea and some strains are enterotoxigenic. Infection has been linked to ankylosing spondylitis as well as myasthenia gravis (antigenic cross-reactivity), and these patients usually carry larger numbers of the organism in their intestines than healthy individuals. Klebsiella oxytoca has been found to be the cause of antibiotic-associated hemorrhagic colitis. These strains have been shown to produce a cytotoxin that is capable of inducing cell death in various epithelial-cell cultures.

Klebsiella is also an infamously known nosocomial infectious agent, partially due to the ability of organisms to spread rapidly. Klebsiella accounts for approximately 3-7% of all hospital-acquired infections, placing it among the top eight pathogens in hospitals. Extraintestinal infection typically involves the respiratory or urinary tracts, but may infect other areas such as the biliary tract and surgical wound sites. K. pneumoniae and K. oxytoca are the two members of this genus responsible for most extraintestinal human infections.

Treatment of these species has become a major problem in most hospitals because of resistance to

multiple antibiotics and potential transfer of plasmids to other organisms. Proper hand washing is crucial to prevent transmission from patient to patient via medical personnel. Contact isolation should be used for patients colonized or infected with highly antibiotic-resistant *Klebsiella* strains.

Klebsiella ozaenae and *Klebsiella rhinoscleromatis* are infrequent isolates that are subspecies of *K. pneumoniae*; however, each is associated with a unique spectrum of disease. *K. ozaenae* is associated with atrophic rhinitis, a condition called ozena, and purulent infections of the nasal mucous membranes. *K. rhinoscleromatis* causes the granulomatous disease rhinoscleroma, an infection of the respiratory mucosa, oropharynx, nose, and paranasal sinuses.

For the otherwise healthy individual, antimicrobial therapy is often unnecessary. *Klebsiella* thrives on a diet high in starch, so a low-starch diet may be helpful. A further caution is that *Klebsiella* thrives on Fructooligosaccharides (FOS) a class of oligosaccharides used as an artificial or alternative sweetener. Antibiotics may be indicated if symptoms are prolonged and in systemic infections. Refer to the bacterial sensitivities to identify the most appropriate pharmaceutical or natural agent.

Hogenauer C, Langner C, Beubler E, et al. *Klebsiella oxytoca* as a Causative Organism of Antibiotic-Associated Hemorrhagic Colitis. *New England Journal of Medicine*. December 2006;355:23.

Levy I et al. Nosocomial Infections After Cardiac Surgery in Infants and Children: Incidence and Risk Factors. *J Hosp Infect*. 2003;53(2):111-6.

Washington W, Allen S, Janda W, Koneman E, Procop G, Schreckenberger P, Woods, G. *Koneman's Color Atlas and Textbook of Diagnostic Microbiology*, 6th edition. Lippincott Williams and Wilkins; 2006. pg 259-264.

Murray PR, Baron EJ, Jorgensen JH, Pfaller MA, Tenover FC, Tenover MC. *Manual of Clinical Microbiology*, 8th edition. Washington, DC: ASM Press; 2003. pg 688-689.

Proteus species

Proteus species are gram-negative bacteria that belong to the Enterobacteriaceae family. *Proteus* species are normal inhabitants of human flora but are listed as dysbiotic flora at 3 - 4+. *P. vulgaris* and *P. mirabilis* are the best-known species of the genus *Proteus*, with *P. mirabilis* being the most common infectious species. *Proteus* is commonly found in contaminated meat, soil, polluted water, sewage and manure, and in the intestinal contents of wild and domestic animals. This bacterium is often found in higher numbers in those who have received antibiotics.

Proteus can produce an enterotoxin, and organisms often localize in already damaged tissues in various parts of the body where they may produce an exudative inflammatory reaction. *Proteus* has been associated with diarrheal disease in susceptible individuals and has been implicated as a possible cause of epidemic diarrhea in infants. In addition, *Proteus mirabilis* is a urease-producing organism that contributes to the formation of kidney stones. Urinary tract infections, most common in patients with long-term catheterizations, may lead to kidney failure and may be the source of bacteremia. Infection has been linked to rheumatoid arthritis (antigenic cross-reactivity). There is

also evidence for osteomyelitis caused by *Proteus* bacteria.

For the otherwise healthy individual, antimicrobial therapy is often unnecessary. Antibiotics may be indicated if symptoms are prolonged and in systemic infections. Refer to the bacterial sensitivities to identify the most appropriate pharmaceutical or natural agent.

Hayward NJ, Jeavons TH, Nicholson AJ, Thornton AG. Methyl mercaptan and dimethyl disulfide production from methionine by *Proteus* species detected by head-space gas-liquid chromatography. *J Clin Microbiol*, 6(3):187-194, 1977.

Jansen AM, Lockett CV, Johnson DE, HT Mobley. Visualization of *Proteus mirabilis* Morphotypes in the Urinary Tract: the Elongated Swarmer Cell Is Rarely Observed in Ascending Urinary Tract Infection. *Infect Immun*, 71(6): 3607-3613, 2003.

Washington W, Allen S, Janda W, Koneman E, Procop G, Schreckenberger P, Woods, G. Koneman's Color Atlas and Textbook of Diagnostic Microbiology, 6th edition. Lippincott Williams and Wilkins; 2006. pg 266-268.

Murray PR, Baron EJ, Jorgensen JH, Pfaller MA, Tenover FC, Tenover MC. Manual of Clinical Microbiology, 8th edition. Washington, DC: ASM Press; 2003. pg 688.

Salmonella group

Salmonella, a gram-negative rod and member of the Enterobacteriaceae family, is listed as dysbiotic in any amount. The bacteria may be isolated from a wide range of mammals, birds, reptiles, human or animal excrement and contaminated water. The salmonellae are the most diverse of all the Enterobacteriaceae, with more than 2,500 serotypes. *Salmonella* is responsible for foodborne toxic infections due to contaminated raw or undercooked meats, eggs, milk products, and plants used as food. This bacterium may also be found in fish, chocolate, dried pasta, and mayonnaise. Prevention of *Salmonella* infection relies on avoiding contamination by improving hygiene and food handling practices. *Salmonella* infections have more severe effects on infants, the elderly, and immunocompromised individuals.

Salmonella can cause either acute gastroenteritis or typhoid fever (typhus). Typhus results from bacterial invasion of the bloodstream. Typically, symptoms of enteritis appear about 12-72 hours following ingestion of contaminated food and last 4-7 days. Symptoms include abdominal pain, diarrhea, vomiting, fever, and possible bloody stools. Most persons recover without treatment. However, in some persons, the diarrhea may be so severe that the patient needs to be hospitalized. In these patients, the infection may cause localized infections (e.g., osteomyelitis or urinary tract infection) or bacteremia. The elderly, infants, and those with impaired immune systems are more likely to have a severe illness. *Salmonella* has also been associated with reactive arthritis. Typhoid fever is a serious bloodstream infection and typically presents with a sustained debilitating high fever and headache, without diarrhea. About 5% of patients clinically cured of *S. typhi* remain carriers for up to one year.

Treatment of enteritis with antibiotics is not advised because it is ineffective, increases the chance of a carrier state developing, and may promulgate the further spread of antibiotic resistance.

Antibiotics may be indicated if symptoms are prolonged and in systemic infections. Refer to the bacterial sensitivities to identify the most appropriate pharmaceutical or natural agent.

CDC Division of Foodborne, Bacterial, and Mycotic Diseases (DFBMD). Available at: http://www.cdc.gov/nczved/dfbmd/disease_listing/stec_gi.html. Accessed October 25, 2009.

Washington W, Allen S, Janda W, Koneman E, Procop G, Schreckenberger P, Woods, G. Koneman's Color Atlas and Textbook of Diagnostic Microbiology, 6th edition. Lippincott Williams and Wilkins; 2006. pg 251-254.

Murray PR, Baron EJ, Jorgensen JH, Pfaller MA, Tenover FC, Tenover FC. Manual of Clinical Microbiology, 8th edition. Washington, DC: ASM Press; 2003. pg 663-665.

Cultured Yeast

Yeast, such as *Candida* are normally present in the GI tract in very small amounts. Many species of yeast exist and are commensal; however, they are always poised to create opportunistic infections and have detrimental effects throughout the body. Factors that contribute to a proliferation of yeast include frequent use of wide-spread antibiotics/low levels of beneficial flora, oral contraceptives, pregnancy, cortisone and other immunosuppressant drugs, weak immune system/low levels of sIgA, high-sugar diet, and high stress levels.

When investigating the presence of yeast, disparity may exist between culturing and microscopic examination. Yeast grows in colonies and is typically not uniformly dispersed throughout the stool. This may lead to undetectable or low levels of yeast identified by microscopy, despite a cultured amount of yeast. Conversely, microscopic examination may reveal a significant amount of yeast present, but no yeast cultured. Yeast does not always survive transit through the intestines rendering it unviable for culturing. Therefore, both microscopic examination and culture are helpful in determining if abnormally high levels of yeast are present.

Microscopic yeast

Microscopic examination has revealed yeast in this stool sample. The microscopic finding of yeast in the stool is helpful in identifying whether the proliferation of fungi, such as *Candida albicans*, is present. Yeast is normally found in very small amounts in a healthy intestinal tract. While small quantities of yeast (reported as none or rare) may be normal, yeast observed in higher amounts (few, moderate to many) is considered abnormal.

An overgrowth of intestinal yeast is prohibited by beneficial flora, intestinal immune defense (secretory IgA), and intestinal pH. Beneficial bacteria, such as *Lactobacillus* colonize in the intestines and create an environment unsuitable for yeast by producing acids, such as lactic acid, which lowers intestinal pH. Also, *Lactobacillus* is capable of releasing antagonistic substances such as hydrogen peroxide, lactocidin, lactobacillin, and acidolin.

Many factors can lead to an overgrowth of yeast including frequent use of antibiotics (leading to insufficient beneficial bacteria), synthetic corticosteroids, oral contraceptives,

and diets high in sugar. Although there is a wide range of symptoms which can result from intestinal yeast overgrowth, some of the most common include brain fog, fatigue, recurring vaginal or bladder infections, sensitivity to smells (perfumes, chemicals, environment), mood swings/depression, sugar and carbohydrate cravings, gas/bloating, and constipation or loose stools.

A positive yeast culture (mycology) and sensitivity to prescriptive and natural agents is helpful in determining which anti-fungal agents to use as part of a therapeutic treatment plan for chronic colonic yeast. However, yeast are colonizers and do not appear to be dispersed uniformly throughout the stool. Yeast may therefore be observed microscopically, but not grow out on culture even when collected from the same bowel movement.

Fecal Lactoferrin

The level of fecal lactoferrin, a biomarker of serious gastrointestinal inflammation, is abnormally high in this fecal sample. Fecal lactoferrin is elevated in association with Inflammatory Bowel Disease (IBD) such as Ulcerative Colitis (UC) or Crohn's Disease (CD)[1,2], but NOT Irritable Bowel Syndrome (IBS)[1,3]. Therefore, assessment of fecal lactoferrin levels enables distinction between IBD and non-inflammatory IBS. Such distinction is critical because, although both IBD and IBS may share some common symptoms such as diarrhea, abdominal cramping and weight loss, the diseases are treated quite differently. IBD may become life threatening, requires life long treatment and possibly surgery. In contrast, IBS is often effectively treated with dietary restrictions, stress reduction and medication.

Gastrointestinal inflammation associated with IBD is associated with increased infiltration of activated neutrophils into the mucosa and increased release of lactoferrin into the gut[1,4,5]. Patients with inflammation of the GI tract, such as IBD (but not IBS), exhibit elevated lactoferrin concentrations in the feces[1].

Clinical studies have shown that fecal lactoferrin levels of healthy persons are similar to IBS patients, but markedly increased in patients with active IBD[1,3]. Patients with IBD oscillate between active and inactive disease states, and fecal lactoferrin levels increase 2-3 weeks prior to onset of clinical symptoms[6]. During remission and effective treatment, fecal lactoferrin decreases significantly. Therefore disease activity, and efficacy of treatment can be monitored by following fecal lactoferrin levels. The test can be ordered separately to track disease activity in patients with IBD.

Moderately elevated levels of fecal lactoferrin can occur, with fecal red blood cells and leukocytes, in association with invasive enteropathogens [7,8]. Such levels would be expected to be much lower than those associated with the active phase of IBD. Therefore, with moderately elevated levels of fecal lactoferrin, one should check for the presence of enteropathogens (eg. Shigella, Campylobacter, Clostridium difficile).

Guidelines for interpreting the results of this test are provided by the results of a large multi-center clinical study which evaluated fecal lactoferrin levels in 180 patients suffering with IBS and IBD (UC and CD) compared to 56 healthy controls.

GROUP	# of SPECIMENS	FECAL LACTOFERRIN
		mean mcg/ml +/- SE
Inactive UC	41	67 +/- 24
Active UC	31	815 +/- 789
Inactive CD	26	239 +/- 83
Active CD	51	672 +/- 242
IBS	31	1.3 +/- 0.3
Healthy Controls	55	1.6 +/- 0.4

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Secretory IgA (sIgA)

The concentration of sIgA is abnormally high in this fecal specimen. Immunological activity in the gastrointestinal tract can be assessed using secretory immunoglobulin A (sIgA). Secretory IgA is the predominant antibody or immune protein the body manufactures and releases in external secretions such as saliva, tears, and milk [1]. It is also transported through the epithelial cells that line the intestines out into the lumen. Secretory IgA represents the first line of defense of the GI mucosa and is central to the normal function of the GI tract as an immune barrier [1]. As the principal immunoglobulin isotype present

in mucosal secretions, sIgA plays an important role in controlling intestinal milieu which is constantly presented with potentially harmful antigens such as pathogenic bacteria, parasites, yeast, viruses, abnormal cell antigens, and allergenic proteins [1]. Secretory IgA antibodies exert their function by binding to antigenic epitopes on the invading microorganism limiting their mobility and adhesion to the epithelium of the mucus membrane [2]. This prevents the antigens from reaching systemic circulation allowing them to be excreted directly in the feces.

Elevated fecal sIgA is an appropriate response to an antigenic presence. Microbial and microscopic studies of the stool are useful in identifying if bacteria, yeast, or parasites are present. Eradication of the pathogenic microorganisms will bring sIgA back down into the normal range. Elevated sIgA levels have been observed in the absence of bacteria, yeast or parasites, in individuals with atopic conditions such as food allergies, urticaria, and dermatitis.

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Short Chain Fatty Acids (SCFAs)

The relative and/or total amounts of SCFAs are abnormal in this specimen. In infants, microbial colonization and subsequent SCFA production is a gradual process which is largely determined by environmental exposure, maternal gut microflora, breastfeeding, and possibly genetics [1]. The establishment of an adequate distribution of healthy flora in the gut is crucial in the health of both infants and adults because of the "competitive exclusion" process of dysbiotic flora [1]. Healthy microflora, such as *Lactobacillus* and *Bifidus* generate large amounts of SCFAs (acetic,

propionic, butyric, and valeric) which decrease the pH of the intestine and therefore make the environment unsuitable for pathogens, including bacteria and yeast [1]. SCFAs are the end product of the bacterial fermentation process of dietary fiber by beneficial flora in the gut and play an important role in the prevention of intestinal dysbiosis [1].

The amount and types of SCFAs produced by colonic bacteria will depend on factors such as type of fiber consumed and overall intestinal health. For example, more SCFAs in general are produced by gums and pectins than oat fiber or corn bran; more propionate and butyrate are produced by bacterial activity on gums than pectins [2]. Antibiotic-induced diarrhea may be secondary to decreased colonic fermentation of carbohydrates and decreased overall production of SCFAs [3].

Studies show that SCFAs have numerous implications in gut physiology and health. SCFAs decrease inflammation, stimulate healing, and contribute to normal cell metabolism and differentiation [4]. Acetate, propionate and butyrate have been demonstrated to potentially improve the microcirculation in the intestinal mucosa thereby improving its growth and repair [5,6]. Rectal irrigation with SCFAs can result in improvement of ulcerative colitis [7]. Butyrate and propionate contribute to apoptosis of colorectal cells by increasing the production of reactive oxygen species in the gut [8]. Butyrate also has a positive effect on the differentiation of colonocytes, and this may account for the protective effect of dietary fiber against colorectal cancer [9]. Probiotics and increased dietary fiber can improve/normalize SCFA status.

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Fecal Calprotectin

The level of fecal Calprotectin is higher than expected. Elevated fecal Calprotectin and Lactoferrin levels indicate the presence of neutrophils and inflammation in the gastrointestinal (GI) mucosa. Calprotectin and Lactoferrin facilitate differentiation between irritable bowel syndrome (IBS) and

inflammatory bowel disease (IBD). IBD includes autoimmune conditions such as Crohns disease and ulcerative colitis (UC); these conditions may become life-threatening and require lifelong treatment.

For patients 4 years old to adults (Manz 2012; Fagerberg 2005):

High levels of Calprotectin ($> 200 \mu\text{g/gm}$) are associated with active IBD and gastrointestinal inflammation; elevation may also occur due to bacterial infection, colitis or sometimes, cancer. Fecal Calprotectin should be reassessed after about 4 weeks for confirmation.

Moderate calprotectin ($50\text{-}200 \mu\text{g/gm}$) is an indicator of chronic inflammation. Inflammation at this level may be due to IBD in remission or inflammation caused by non-steroidal anti-inflammatories (NSAIDs). Levels should be reassessed after about 4 weeks. Low levels of Calprotectin ($< 50 \mu\text{g/gm}$) are usually associated with viral GI infections or non-inflammatory bowel conditions such as IBS.

Multiple studies have shown fecal Calprotectin and Lactoferrin to be equivalent with respect to clinical sensitivity and specificity. Studies suggest that Calprotectin may correlate more closely with histological (cell microscopy) findings. Lactoferrin may correlate better to macroscopic (endoscopy) findings, and may be the better indicator of impending relapse, elevating 2-3 weeks prior to clinical symptoms.

Chronic inflammation of the gastrointestinal mucosa contributes to symptoms of IBD. Chronic stress is known to contribute to symptom flare-ups and increased inflammation. Liver disease or the use of aspirin or nonsteroidal anti-inflammatory (NSAID) medications may elevate Calprotectin levels. Fecal Calprotectin levels may also be increased in newborns.

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